POLICY AND ENGINEERING PROPOSAL | ACN-01 / DRAFT 1.0

# Standardizing an American Canal Network

A U.S. Department of Transportation-style report for a hypothetical national freight and passenger canal system



## One common vessel envelope. Reliable clearances. Published operating limits.

Proposed standards for channels, bridges, locks, passenger craft and intermodal terminals, informed by the broad American canal, grain-barge and three-level luxury cruiser concepts developed in this design series.

| BASELINE DESIGN | PROPOSED VALUE |
|---|---|
| Navigation bed width / maintained depth | 70 m / 4.5 m (230 ft / 14.8 ft) |
| Usable lock chamber | 110 m x 18 m (361 ft x 59.1 ft) |
| Bridge vertical clearance | 12 m (39.4 ft) at high navigable water |

STATUS: independent policy and engineering concept. No DOT, USACE or USCG sponsorship, adoption, approval, appropriation or route authorization is asserted. Numeric minima below are proposed design requirements, not existing national regulations.

01 / EXECUTIVE RECOMMENDATION

# Adopt a corridor standard, not a nationwide promise

Establish a proposed American Canal Network (ACN) interoperability framework for new controlled-water canals and selected improvements to existing waterways. Its baseline, ACN-100, serves individual grain barges, single-unit container vessels and broad-beam passenger cruisers. The designation refers to a 100 m maximum operating-unit length, not channel width.

| Core decision | Recommendation |
|---|---|
| Design vessel | Up to 80 m long x 15 m beam x 3.0 m draft; up to 10 m air draft. A tow plus tug may reach 100 m overall. |
| Civil works baseline | 70 m bed width; 4.5 m maintained depth at low navigable water; 110 m x 18 m usable locks. |
| Bridge standard | 12 m vertical clearance at high navigable water; 100 m clear horizontal opening at new two-way canal bridges, subject to greater local requirements. |
| Passenger compatibility | A proposed 40 m x 8 m cruiser envelope, draft 1.8 m and air draft 8 m, fits within ACN-100. |
| Existing river freight | Preserve larger existing navigation needs. Provide a separate large-tow overlay rather than reducing waterways to ACN-100. |
| Cross-country links | Rocky Mountain and Arizona images illustrate aspirations only. No continuous transcontinental navigation route is established. |

## Approval sought in this proposal

Authorize preparation of a technical standard and corridor feasibility studies, rather than construction of an entire coast-to-coast canal. Require route-specific navigation simulation, hydrology, environmental review, geotechnical studies and public engagement before any capital commitment.

In this document, “shall” means a proposed contractual requirement if adopted. It does not create a legal obligation today. SI dimensions control this draft; U.S. customary conversions are rounded for reference.

02 / NATIONAL CONTEXT AND RESPONSIBILITIES

# Build on U.S. navigation practice

U.S. inland waterways already support large commercial tows. USACE identifies the common jumbo barge as 195 ft x 35 ft (59.4 m x 10.7 m). Smithland has 1,200 ft x 110 ft locks, while the Tennessee-Tombigbee Waterway uses 600 ft x 110 ft locks and a 9 ft navigation depth. These examples inform interoperability; they are not the proposed ACN baseline. [1-3]

| Institution | Proposed role; existing authority remains |
|---|---|
| USDOT / MARAD | Coordinate a proposed national interoperability policy and intermodal case. Marine Highway designation does not itself build or operate a canal. [4] |
| USACE | Navigation engineering, relevant federal civil-works responsibilities and applicable permitting / alteration review. [5] |
| USCG | Bridge navigation review and applicable vessel / waterway safety functions. Bridge clearances depend on reasonable navigation needs at the site. [6-7] |
| States, Tribes and local bodies | Land and water interests, consultation, approvals, emergency response and local infrastructure within their respective authority. |
| Canal asset owner / operator | Maintain the advertised navigation envelope, operate locks, issue restrictions and publish verified route data. |
| Vessel owner / master | Supply actual dimensions, maintain a safe vessel and plan each voyage against current restrictions. |

## A national standard is an interface agreement

It makes vessel dimensions, operating water levels and clearances comparable. It cannot override bridge permits, water rights, environmental obligations, Tribal interests, navigation rules or the authority of an existing waterway operator. Larger existing bridge clearances shall not be reduced to meet a smaller ACN minimum.

03 / REFERENCE FLEET

# Define the craft before the infrastructure

| Proposed fleet envelope | Length | Beam | Draft | Air draft |
|---|---|---|---|---|
| P-40 passenger / liveaboard | 40 m / 131 ft | 8 m / 26.2 ft | 1.8 m / 5.9 ft | 8 m / 26.2 ft |
| F-80 self-propelled freight | 80 m / 262 ft | 15 m / 49.2 ft | 3.0 m / 9.8 ft | 10 m / 32.8 ft |
| T-100 coupled operating unit | 100 m / 328 ft | 15 m / 49.2 ft | 3.0 m / 9.8 ft | 10 m / 32.8 ft |

The earlier cargo image suggested a roughly 65 m x 11 m vessel. F-80 provides room for a larger standardized successor. The three-level luxury concept suggested a roughly 30 m broad-beam cruiser. P-40 adds a planning envelope for that family of craft. None of the generated images is a measured vessel design.

## Envelope rules

Length is extreme operating length including pushboat, couplings and projecting gear. Beam includes fixed projections, fenders and cargo overhang. Air draft is height above the actual waterline to the highest item that remains raised in transit. Draft is the deepest immersed point in the actual loading condition.

A 100 m tow shall fit as a complete operating unit; it is not permission for a 100 m barge plus an additional tug. No multi-barge, three-wide river tow is implied by T-100. Passenger certificates, stability, fire protection, lifesaving and access requirements remain separate from dimensional fit.

## Load condition matters

Check a deeply loaded vessel for bottom clearance and a lightly loaded vessel for overhead clearance. Maximum permitted draft and maximum air draft need not occur together. Containers, cranes, roof terraces, antennas, awnings and solar equipment all count. Do not rely on ballast or loading a vessel down merely to clear a bridge.

04 / MASTER DIMENSIONAL SCHEDULE

# ACN-100: the common civil-works baseline

| Feature | Proposed minimum / limit | Measurement basis |
|---|---|---|
| Reference operating unit | 100 m length; 15 m beam | Extreme operating envelope, including tug |
| Permitted design draft | 3.0 m / 9.8 ft | Actual draft; restrictions may reduce it |
| Permitted design air draft | 10.0 m / 32.8 ft | Highest transit item above actual waterline |
| Channel maintained depth | 4.5 m / 14.8 ft | At declared low navigable water (LNW) |
| Straight two-way bed width | 70 m / 230 ft | Width of maintained-depth navigation prism |
| Usable lock chamber | 110 m x 18 m / 361 ft x 59.1 ft | Clear working area between obstructions |
| Water depth over lock sill | 4.5 m / 14.8 ft | At lowest authorized chamber / approach level |
| Bridge vertical opening | 12 m / 39.4 ft | Lowest obstruction above high navigable water (HNW) |
| New two-way bridge width | 100 m / 328 ft | Clear between fenders; greater if required |
| Bank / lock coping freeboard | 1.5 m / 4.9 ft above HNW | Planning floor; increase for floods, waves and settlement |
| Routine planning lift per lock | 6 m / 19.7 ft | Illustrative module; not a universal safe lift |
| Passenger reference craft | 40 m x 8 m; 8 m air draft | Allows main deck, upper deck and lower cabins |

## Application

Minima apply to proposed ACN-100 new construction. Site design may increase them, and the relevant authority may impose a different opening. Bends, passing reaches, bridge approaches, waves, crosswinds and currents require simulation. Reduced dimensions require a published restriction and explicit class downgrade, not an undisclosed exception.

05 / WATER LEVELS AND BRIDGE CLEARANCES

# Measure headroom from the right water level

Each reach shall publish LNW, normal target level and HNW in a surveyed vertical datum, together with the local gauge reference and closure thresholds. LNW and HNW are declared navigation limits, not synonyms for every flood or drought event. At tidal interfaces, reconcile the navigation datum with the reference level required by the bridge authority. [6-7]

Fixed bridge requirement: lowest soffit, service pipe, inspection traveler or other obstruction shall be at least 12 m above HNW throughout the approved navigation opening. Do not measure only at the center if an arch or girder limits the sides. Lights, signs, cables and maintenance equipment shall not intrude into the declared box.

## Worked elevation example

| Item | Example elevation / result |
|---|---|
| LNW | 100.0 m in a hypothetical local datum |
| HNW | 101.0 m in that same datum |
| Lowest bridge obstruction | At least 113.0 m |
| Clearance at HNW | 113.0 - 101.0 = 12.0 m |
| Clearance above 10 m transit air draft | 2.0 m static allowance at HNW |

The 2 m allowance is not permission to consume all margin in waves or motion. The master and operator must allow for heel, trim, heave, uncertainty and operating conditions. Suspend passage where the required residual margin cannot be maintained.

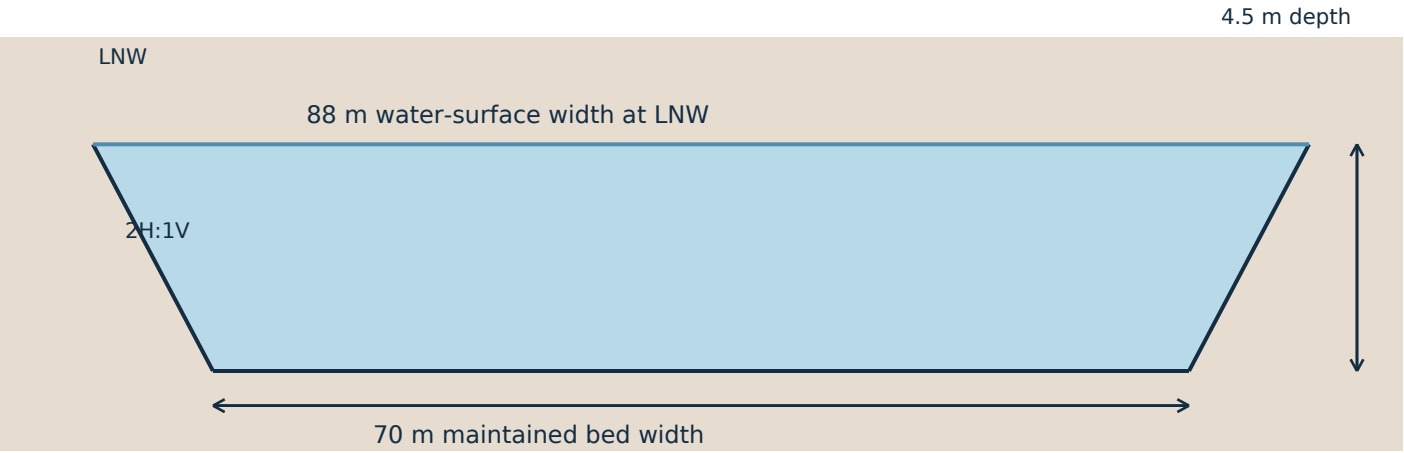
## Movable bridges and utility crossings

Where a fixed crossing cannot provide the envelope, a movable bridge may qualify only with an approved open-position navigation box, reliable opening procedures and published outages. Overhead utilities need electrical and other clearances beyond this navigation minimum; adopt the greater applicable requirement. USCG determines bridge navigation clearances case by case; ACN targets do not replace its process. [6-7]

06 / CHANNEL GEOMETRY

# Keep width, depth and bank profile distinct

The 70 m baseline is bed width at maintained depth, not the total water-surface width. The sketch below shows one possible rural excavation; retaining walls may be used at urban sites subject to wake, reflection and structural design.




Schematic cross-section; horizontal and vertical scales differ. Dimensions are proposed planning values, not a bank-stability design.

At a 2 horizontal : 1 vertical bank slope, surface width at LNW is 70 + 2 x 2 x 4.5 = 88 m. At a level 1 m above LNW, it becomes 92 m. The proposed 100 m bridge opening gives space for this example geometry; fenders and foundations must still preserve the navigation prism.

## Provisional depth budget

| Draft | Net clearance | Squat allowance | Survey allowance | Shoaling reserve | Total |
|---|---|---|---|---|---|
| 3.0 m | 0.5 m | 0.3 m | 0.2 m | 0.5 m | 4.5 m |

This additive planning budget is provisional. Establish squat, trim, bed tolerance and under-keel clearance through vessel and reach analysis without double-counting allowances. Construction overdepth is extra and does not raise the advertised draft limit.

07 / NAVIGATION AND BRIDGE APPROACHES

# Straight-reach minima are not a design shortcut

Validate the 70 m bed width using simulations of two F-80 vessels meeting under the limiting wind, current, bank effect and water depth. Cross-sections shall include a controlled navigation prism, bank protection, maintenance access and any separate ecological shelf. USACE shallow-draft waterway guidance provides a relevant design framework; the ACN numbers remain proposed inputs. [8]

| Location | Proposed treatment |
|---|---|
| Bends and junctions | Set bend radius and widening by swept-path and shiphandling analysis; do not impose a single national radius without a design vessel and operating speed. |
| Narrow cutting or tunnel | Default to one-way controlled working unless full two-way clearance is demonstrated. Publish ventilation, evacuation, draft and air-draft limits. |
| Lock approaches | Provide separate arrival holding and departure clearance; protect vessels from approach currents and gate discharge. |
| Passenger marinas | Use off-channel basins outside freight swept paths, with wake control and safe pedestrian access. |
| Public towpaths | Separate public access from mooring machinery, maintenance traffic and cargo operations. |
| Ecological shelves | Place habitat zones outside the maintained navigation prism; do not count their shallow water toward required width. |

## Operating speed

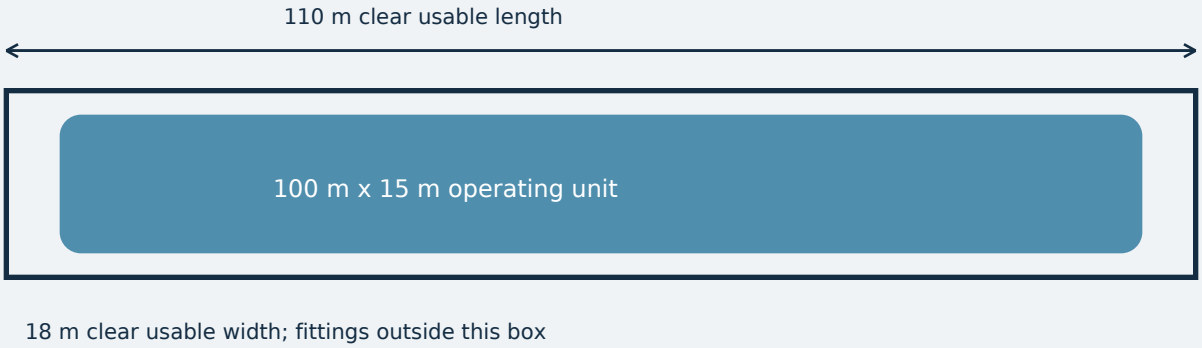
Set reach-specific speeds by wake, bank erosion, stopping distance and interaction testing. There is no universal cruise speed or coast-to-coast journey time in this proposal. Conditions may require no passing, escort, daylight-only transit or closure. Publish these rules in the route register and communicate changes before entry.

Capacity planning shall include approach queues, bridge openings, maintenance windows, climate disruption and passenger use; a nominal channel width alone does not establish throughput.

08 / LOCK DESIGN

# A usable 110 m x 18 m chamber

Usable dimensions are measured between all gate sweeps, fenders, ladders, bollards and other intrusions at the operating condition. Civil wall-to-wall dimensions must be larger where fittings consume space. A 100 m x 15 m operating unit leaves 10 m total end space and 3 m total side space; the final distribution and safe movements need validation.




Plan schematic is not to scale. Gate machinery, wall thickness, culverts and approach works are additional.

| Subsystem | Proposed requirement |
|---|---|
| Sills and approaches | 4.5 m water depth at the lowest permitted level; publish local draft restrictions. |
| Gates and valves | Interlock gate movement with equalized levels, valve state and verified clear chamber; provide independent emergency stops. |
| Mooring | Design floating or appropriately spaced bollards for the full operating range, with accessible passenger-craft positions. |
| Hydraulics | Prove acceptable filling currents, vessel forces and surge using modelling and commissioning tests. |
| Recovery | Provide stoplogs / emergency isolation provisions, drainage, access and tow-out arrangements. |

Any overhead structure across a lock shall preserve 12 m headroom above the highest navigable chamber level. Plan coping at least 1.5 m above HNW, subject to greater flood, wave and settlement allowances. Wall height depends on sill depth, level difference and freeboard; gate height and foundations require separate structural design.

A 6 m planning lift is not proof of identical gates, filling times or foundations at every site. A different lift or boat lift needs separate engineering and approval.

09 / LOCK FLIGHTS AND AUTOMATION

# Controlled steps, not a sloping waterway

A six-chamber flight at 6 m per chamber provides 36 m total rise. Each chamber must have a horizontal water surface, independently controlled level and protected approach. Shared gates in a staircase require a sequence that cannot drain an occupied chamber into an unsafe condition.

| Automatic sequence | Verification before transition |
|---|---|
| 1. Admit | Confirm identity, dimensions, chamber readiness and entry signal. |
| 2. Secure | Detect all vessels inside the safe area; confirm mooring and gate exclusion zones clear. |
| 3. Isolate | Close and confirm entry gate; verify interlocks and level sensors. |
| 4. Fill / empty | Operate valves under monitored rate and vessel-force limits. |
| 5. Equalize | Confirm levels across exit gate are within validated tolerance. |
| 6. Release | Open exit gate, clear departure zone, then reset for the next lockage. |

## Retain human oversight and local control

Remote automation shall not remove an accountable lock operator. Use independent level verification, obstruction detection, authenticated commands, event logs, backup communications and a tested manual operating mode. A communication loss shall enter a defined safe state; it must not trigger an automatic draining or opening sequence.

Lockage time is not established by this draft. Do not infer a commercial transit time from valve filling time alone: vessel entry, mooring, checks, equalization, departure, queues and intermediate pounds all contribute.

## Mountain feasibility

A hypothetical 1,000 m ascent would need about 167 lifts at 6 m, plus any descent on the far side. This arithmetic shows why the Rocky Mountain imagery is not a route design. Water supply, geology, winter conditions, construction access and economics may rule out a crossing entirely.

10 / WATER DEMAND AND ENERGY

# Treat water as a binding design constraint

For the reference lock, gross chamber exchange for a 6 m lift is approximately 110 x 18 x 6 = 11,880 m³ per level change. This is a simplified geometric volume, not a system-wide water-consumption figure. A lock flight, simultaneous traffic and return pumping require a coupled water-balance model.

| Illustrative calculation | Result / interpretation |
|---|---|
| Assumed water-saving recovery | 60% of gross volume; a design scenario, not a guarantee |
| Remaining make-up per level change | 11,880 x 0.40 = 4,752 m³ |
| Pump that volume upward 6 m | E = density x g x volume x head / efficiency |
| At 80% pumping efficiency | About 97 kWh, excluding pipes, auxiliary loads and additional head |
| 100 km rural reach at 88 m surface width | 8.8 million m² water surface at LNW |
| Assume 6 mm/day net evaporation | 52,800 m³/day; climatic sensitivity only, not a measured Arizona rate |

Approve a seasonal and drought water balance before route selection. Include evaporation, seepage, lock operation, leakage, rainfall, ecological flows, storage, salinity and legally available supply. Recirculation reduces some operational withdrawals; it does not eliminate evaporation or supply constraints.

## Arizona and interbasin safeguards

No Colorado River allocation, groundwater entitlement or other new water supply is assumed. Do not connect watersheds merely because the vessel envelope fits. Cross-basin channels can spread invasive species; physical isolation, vessel treatment or a non-water transport transfer may be necessary. [9]

Where water or ecological conditions cannot support navigation, use an intermodal freight transfer or a different corridor. A standard should permit a rational rejection of a route, not require its construction.

11 / PASSENGER CRAFT COMPATIBILITY

# Make room for the three-level cruiser



Illustrative interior concept from the design series. Not a naval-architecture drawing, stability assessment or proof of its dimensions.

P-40 offers a proposed 40 m length, 8 m beam, 1.8 m draft and 8 m transit air-draft envelope. The earlier roughly 30 m cruiser could fit within it only after a naval architect confirms its actual geometry, loading, structure and stability. Count the upper lounge, rails, antennas and terrace fittings in the height assessment.

| Passenger design provision | Proposed treatment |
|---|---|
| Bridge fit | 8 m maximum P-40 air draft versus 12 m infrastructure clearance at HNW; actual voyage margin still checked. |
| Upper floor and roof terrace | Verify wind heel, crowd loading, stability, guardrails and emergency escape; no automatic approval from dimensional fit. |
| Lower-deck cabins | Provide compliant escape and flooding protection; the cutaway image is not a watertight subdivision plan. |
| Passenger lock use | Protected mooring locations, clear passenger instructions and segregation from cargo transfer machinery. |

Certification, inspected-passenger-vessel requirements where applicable, accessibility and accommodation standards require a separate regulatory determination for the vessel and service.

12 / FREIGHT TERMINALS AND ENERGY

# Standardize the interfaces, not the cargo machinery

Provide off-channel freight basins sized for the declared fleet, with approach protection and safe waiting berths. A vessel shall be secured before robotic lifting begins. Dedicated container spreaders shall engage approved corner fittings; bulk grain terminals need commodity-appropriate hoppers and dust controls.

| Terminal interface | Requirement proposed for project specifications |
|---|---|
| Berth and basin | Declare vessel size, berth depth, mooring loads, fender performance and approach restrictions. |
| Robotic cargo handling | Separate moving loads, operators, public access and automated vehicles; validate machine safeguards and emergency stopping. |
| Passenger interface | Separate passenger landings from industrial berth envelopes and vehicle lanes. |
| Shore power | Select applicable electrical standards, connector ratings, protections and isolation arrangements with operator and utility. |
| Low-carbon fuels | Fuel-specific assessment for storage, transfer, ventilation, detection and emergency response; no common berth presumed suitable for all fuels. |
| Waste and services | Provide pump-out and waste reception where required, potable water controls and spill-response equipment. |

The Los Angeles robotic-port concept illustrates a possible terminal function. It is not an approved port expansion, a selected alignment, a demonstration of cargo throughput or evidence of an existing canal into the port.

Do not bake one propulsion system into the civil standard. Hydrogen-electric, battery-electric and other compliant craft may share the route when dimensions, performance and local operational requirements are satisfied. Emissions claims require lifecycle and energy-supply evidence, not a label on the hull.

13 / DIGITAL ROUTE REGISTER

# Publish what a vessel can actually navigate

Create a versioned route register maintained by the asset operator. Every reach, lock, bridge, marina and terminal shall have a unique identifier and a link to the authoritative operating notice. Static engineering values and live operating restrictions must be distinguishable.

| Record group | Minimum proposed fields |
|---|---|
| Spatial and survey | Reach limits; horizontal and vertical datum; survey date; data owner; survey uncertainty. |
| Vessel envelope | Maximum operating length, beam, draft and air draft; any coupled-unit rule. |
| Water and depth | LNW / HNW; gauge reference; current level and timestamp; controlling depth; dredging and shoaling notices. |
| Bridge and lock geometry | Lowest obstruction elevation; clear opening; useful chamber dimensions; sill elevations; movable bridge status. |
| Operations | Speed and passing rules; closures; weather limits; booking requirements; inspection periods. |
| Passenger access | Marina services; accessible facilities; emergency contacts; passenger mooring rules. |
| Data assurance | Version; source; validity period; update timestamp; outage state; audit log. |

A clearance calculator may assist route planning, but shall show its data age and assumptions. Unknown levels or expired surveys shall not silently produce a green “safe to transit” result. Onboard instruments and the master’s responsibility remain necessary.

Cybersecurity shall separate safety-critical lock controls from public booking and payment services. Operators need tested backups, access control, incident response and local recovery. No vendor-specific protocol or assumed autonomous-vessel approval is established here.

14 / EXISTING WATERWAYS AND LARGER TOWS

# Interoperability without forced reconstruction

Publish three statuses: ACN-100 compliant; restricted connector with an explicit smaller envelope; and large-tow overlay with larger route-specific dimensions. Existing waterways do not become noncompliant with law because they differ from this proposed scheme.

| Network layer | Planning interpretation |
|---|---|
| ACN-100 new controlled canal | Baseline 110 m x 18 m lock, 70 m bed and 12 m bridge clearance. Two-way operation must be demonstrated. |
| Restricted existing connector | Advertise actual limits; a 9 ft project depth does not support the 3 m ACN draft by default. Route eligibility depends on under-keel clearance. |
| Large-tow overlay | Use a planning reference of 366 m x 33.53 m (about 1,200 ft x 110 ft) usable lock, informed by established USACE facilities. [2] |
| Large-tow channels / bridges | Size to actual tow and navigation needs. No 70 m channel or 12 m headroom cap applies automatically. |
| Missing or unsuitable link | Use rail, road or coastal shipping interchange; do not represent a physically disconnected system as through navigation. |

## Why the layers matter

An 18 m-wide baseline chamber can pass a 10.7 m-wide standard jumbo barge individually, but cannot pass a three-wide 32 m tow. Large tows require separate geometry, breakup operations or transfer. A 600 ft x 110 ft existing lock is wider and longer than ACN-100, but its project depth and bridges may still control access. [1-3]

Port interfaces may require higher clearances and deeper channels than ACN-100. Maintain the greater existing or authority-approved navigation requirement. Adoption shall not create a new bottleneck for established traffic.

15 / APPROVAL AND DELIVERY PATHWAY

# Advance in stages with stop decisions

| Stage | Evidence before progression |
|---|---|
| A. Standard development | Multi-agency working group; vessel and cargo demand; provisional dimensions; accessibility and environmental input. |
| B. Corridor screening | Existing-waterway option first; rail / road / coastal alternatives; water availability; geology; land and Tribal interests. |
| C. Feasibility | Navigation simulation; hydrology; geotechnical investigation; ecosystem and community impacts; whole-life economic comparison. |
| D. Design and approvals | Applicable federal and state processes; bridge review; permits; water rights; land acquisition; funding and maintenance commitment. |
| E. Demonstration | One representative lock / bridge / terminal interface; instrumented shiphandling and emergency tests; independent acceptance. |
| F. Phased construction | Deliver useful sections with operating connections; publish restrictions until all connecting assets meet the advertised class. |
| G. Operating assurance | Surveys, inspections, dredging, gate tests, incident review and funded renewal. |

Applicable U.S. processes shall be identified by counsel and the responsible agencies. USACE administers relevant Section 10 and Section 404 permitting, and alterations to existing USACE projects can require Section 408 review. Bridge review is separate. Broader environmental review and other approvals depend on the project and current law. [5-7]

No capital cost, toll revenue, benefit-cost ratio, completion date or national route mileage is claimed. These require actual alignments, excavated quantities, structures, property costs, water and energy plans, demand, environmental measures and maintenance obligations. A PPP does not remove these requirements.

16 / ACCEPTANCE AND COMPLIANCE

# Verify the advertised class before opening

| Acceptance item | Required evidence / response |
|---|---|
| As-built geometry | Survey clear navigation prism, chamber useful dimensions, sill levels and bridge obstructions in the declared datum. |
| Vessel interaction | Full-mission simulation, representative commissioning transits and operational limits for meetings / bends. |
| Water and hydraulics | Demonstrate seasonal supply and validated lock filling / emptying without unsafe vessel loads. |
| Safety and recovery | Test gate / valve interlocks, safe states, emergency isolation, tow-out and emergency response. |
| Digital publication | Verify route register values, timestamps, closure notices and consistency with signs and gauges. |
| Maintenance funding | Adopt inspection, dredging, gate overhaul and asset-renewal plan before service opens. |
| Independent review | Competent engineering acceptance plus relevant authority approvals; no self-certified national badge. |

## Deviation process

Every departure shall identify the affected vessel envelope, the reason, risk controls, responsible approver, expiry and public notice. A temporary reduction in depth may impose a draft restriction; it does not change the original as-built depth. A permanent smaller opening means route reclassification unless an approved operating solution preserves the envelope.

Annual public reporting should include controlling depths, closures, incidents, maintenance backlog, water use, energy demand, passenger access and commodity movements. Do not use gross theoretical lock capacity as measured freight throughput.

## Recommended adoption resolution

Adopt ACN-01 as a draft planning framework for consultation and testing. Reserve final dimensional approval to the responsible engineering and permitting authorities. Authorize no mountain or desert link until it independently passes feasibility, water, environmental and economic gates.

APPENDIX A / DIMENSIONAL CHECKS

# Transparent calculations and assumptions

| Check | Calculation | Result |
|---|---|---|
| Baseline vessel / lock length | 110 m - 100 m operating unit | 10 m total end allowance |
| Baseline vessel / lock beam | 18 m - 15 m extreme beam | 3 m total side allowance |
| Bridge / freight headroom | 12 m - 10 m air draft | 2 m static allowance at HNW |
| Bridge / passenger headroom | 12 m - 8 m air draft | 4 m static allowance at HNW |
| Reference rural surface width | 70 + 2 x 2 x 4.5 | 88 m at LNW |
| Same width at HNW = LNW + 1 m | 70 + 2 x 2 x 5.5 | 92 m; survey actual crossing |
| Six-lock flight | 6 locks x 6 m lift | 36 m rise |
| Gross level-change volume | 110 x 18 x 6 | 11,880 m³ |
| 60% recovery scenario | 11,880 x (1 - 0.60) | 4,752 m³ make-up |
| Example pumping energy | 1,000 x 9.81 x 4,752 x 6 / (0.8 x 3,600,000) | 97.1 kWh |
| Evaporation sensitivity | 100,000 x 88 x 0.006 | 52,800 m³/day |

## Boundaries of the calculations

Allowances shown are not validated dynamic safety margins. Lock volume ignores detailed chamber geometry and vessel displacement effects. Recovery is hypothetical; pumping energy uses a 6 m lift only, not an entire flight or watershed. Evaporation is an assumed sensitivity at 6 mm/day with no rainfall offset, not a regional climate finding.

Key conversions: 1 m = 3.28084 ft; 3 m = 9.84 ft; 4.5 m = 14.76 ft; 12 m = 39.37 ft; 18 m = 59.06 ft; 70 m = 229.66 ft; 100 m = 328.08 ft; 110 m = 360.89 ft. Do not round the design requirement down when procuring in U.S. customary units.

APPENDIX B / PRIMARY REFERENCES

# Published context and design guidance

Sources consulted 8 November 2026. These sources support factual context and process descriptions. None adopts the proposed ACN dimensions or endorses the speculative route imagery. Source numbering is used in the report text.

[1] USACE, EM 1110-2-3402: Barge Impact Forces for Hydraulic Structures (2022), Appendix D
Open primary source - Common jumbo-barge dimensions and the relationship between flotilla configuration and lock design.

[2] USACE, Smithland Locks and Dam
Open primary source - Published 1,200 ft x 110 ft chambers; an existing large-lock comparator.

[3] USACE, Tennessee-Tombigbee Waterway: Navigation
Open primary source - Published 600 ft x 110 ft locks and 9 ft navigation depth; regional comparator.

[4] MARAD, U.S. Marine Highway Route Designation Process
Open primary source - Role of USDOT route designation; it is not authorization for this hypothetical canal.

[5] USACE, Jurisdiction and Permit Authorities
Open primary source - Section 10, Section 404 and Section 408 context; actual applicability is project-specific.

[6] USCG, Bridge Permit Application Guide (July 2025)
Open primary source - Navigation opening, lowest obstruction and appropriate high-water reference documentation.

[7] USCG, Bridge Guide Clearances
Open primary source - Reasonable navigation needs and case-by-case clearance determination.

[8] USACE Publications, Engineer Manuals catalogue: EM 1110-2-1611
Open primary source - Layout and Design of Shallow-Draft Waterways; relevant guidance to consult in full for actual design.

[9] EPA Watershed Academy, Invasive Non-Native Species
Open primary source - Cross-basin canals and transfers as potential pathways for invasive species.

Image credits: AI-generated American Canal Network design studies developed in this conversation. All schematics in this report were drawn separately for the proposed dimensional framework; they are not surveyed construction drawings.